

MB Problems: Electric Charges and Electric Fields

Answers to MB problems are on page M-4. **Updated Monday September 7, 2026 7:01pm.**

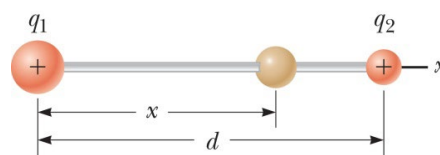
MB1. Richard Feynman, a Nobel Prize-winning physicist, once noted that if two people stood an arm’s length away from each other, and each person had just 1% more electrons than protons, the force pushing them away from each other would be enough to lift the “weight” of the entire planet Earth. Do an order-of-magnitude calculation to determine whether this is true. (*Hint: the fact that he won a Nobel Prize in physics should be your clue that it’s probably true.*)

MB2. Caitlin Clark and the Indiana Fever are playing two games against A’ja Wilson and the Las Vegas Aces. Suppose that there are two basketballs, one in Indianapolis and one in Las Vegas, and suppose that by some miracle, the basketball in Indianapolis suddenly loses all of its electrons, and the one in Las Vegas suddenly loses all of its protons. (We’ll ignore for the moment that if this actually happened, each ball would instantly blow itself to smithereens.) Do an order-of-magnitude calculation to show that the attractive force between the two balls at this distance would be so great that all of the members of both teams working together wouldn’t be strong enough to hold them apart.

MB3. The gravitational force between two objects of mass m_1 and m_2 is given by $F_G = \frac{Gm_1m_2}{r^2}$, where the gravitational constant $G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$. In a hydrogen atom, the electron and the proton are separated by an average distance of one Bohr radius, or about $5 \times 10^{-11} \text{ m}$. Which force is stronger between the hydrogen’s electron and the proton: the electrical force or the gravitational force?

MB4. Two small balls with positive charges $q_1 = 3q$ and $q_2 = q$ are glued to the opposite ends of a horizontal insulating rod of length $d = 1.50 \text{ m}$. The q_1 ball is at the origin. A third small ball, also charged, is free to slide back and forth on the rod.

- (a) At what x position is the third ball at equilibrium?
 (b) Can that equilibrium be a stable equilibrium? Explain.

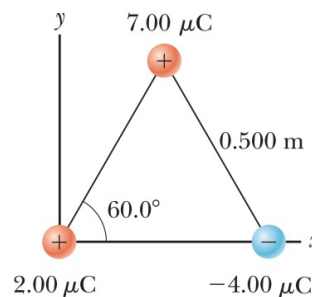


MB5. Three charged balls are on the corners of an equilateral triangle, as shown. What is the total electric force on the ball with the $7.00 \mu\text{C}$ charge?

MB6. A regulation NHL hockey puck measures three inches across and one inch thick ($R = 3.81 \text{ cm}$, $h = 2.54 \text{ cm}$). Suppose that a particular puck carries a charge density ρ that varies with the distance r from the center of the puck¹ according to

$$\rho = (430 \text{ nC/m}^2) \left(\frac{1}{r} \right) \sin \left(\pi \frac{r}{R} \right).$$

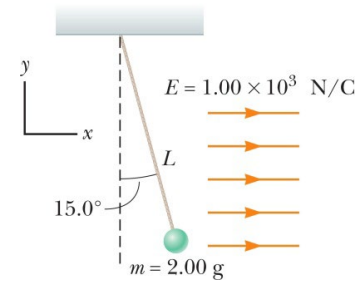
What is the total charge Q on the puck?



MB7. Consider the problem that we did in the lab, where we found the electric field near a charged rod. Suppose you cut the rod in half and threw away the bottom part, leaving only a rod of charge $Q/2$ extending from $y = 0$ to $y = L/2$. Find the x and y components of the electric field at the point P .

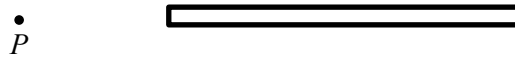
¹There is no physical reason to expect this particular charge distribution on a hockey puck in any league. It’s just a fictional problem based on an arbitrary function that I think you can integrate.

MB8. A tiny ball with mass $m = 2.00$ grams hangs by a 20-cm-long string in a uniform electric field of 1000 N/C. (You don’t need to worry about where the field comes from—presumably it comes from some other charges not pictured.) What would the net charge on the ball need to be so that it is in equilibrium when the string makes a 15.0° angle with the vertical?

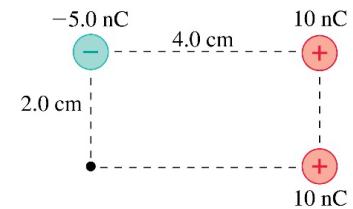


MB9. Consider the problem from Lab 6 where you found the electric field $\vec{E}$ a distance x from a charged rod. (a) Express your final answer for $\vec{E}$ in terms of the charge density λ per unit length, rather than Q and L . (b) What is the electric field a distance x from an *infinitely long* rod with uniform charge density λ ? That is, what is your answer for part (a) in the limit of $L \rightarrow \infty$?

MB10. A thin rod has charge Q and length L . Find the electric field $\vec{E}$ at a point P that lies along the rod’s axis and is a distance d from the end of the rod. (*Hint: put the rod along the x axis with point P at the origin.*)



MB11. (From Knight 4e, 23.36) What are the strength and direction of the electric field at the position indicated by the dot in the figure? Give your answer (a) in component form, and (b) as a magnitude and angle measured cw or ccw from the positive x axis.



MB12. (From Knight 5e, 22.16) A small glass bead has been charged to $+20$ nC. A small metal ball bearing 1.0 cm above the bead feels a 0.018 N electric force, pulling it downward. What is the charge on the ball bearing?

MB13. (From Knight 5e, 22.29) The electric field at a point in space is $\vec{E} = (400\hat{i} + 100\hat{j})$ N/C.

- What is the electric force on a proton at this point? Give your answer in component form.
- What is the electric force on an electron at this point? Give your answer in component form.
- What is the magnitude of the proton’s acceleration?
- What is the magnitude of the electron’s acceleration?

Answers to Selected MB Problems:

MB2. My estimate for the force between the balls is $\sim 10^{11}$ N. That’s about 10 billion pounds of force! (How many pounds of force do you think you could apply to the ball if you were trying to hold it back? How would your answer change if your name was A’ja Wilson?)

MB3. The electric force is about 2×10^{39} times stronger than the gravitational force!

MB4. (a) 0.951 m (b) Hmmm... depends on the signs of the charges....

MB5. 0.872 N, 30° below the x axis

MB6. 5.23 nC

MB7. $E_x = \frac{k_E Q}{2x} \frac{1}{\sqrt{x^2 + (L/2)^2}}, E_y = -\frac{k_E Q}{L} \left(\frac{1}{x} - \frac{1}{\sqrt{x^2 + (L/2)^2}} \right)$

MB8. $5.25 \mu\text{C}$

MB9. (a) $E = \frac{k_e \lambda}{x \sqrt{(x/L)^2 + (1/2)^2}}$ (b) $E = \frac{2\pi k_e}{x}$

MB10. $E = \frac{k_e Q}{d(d+L)}$

MB11. (a) $(-9.7\hat{i} + 9.2\hat{j}) \times 10^4 \text{ N/C}$

MB12. 10 nC

MB13. (a) $(6.4\hat{i} + 1.6\hat{j}) \times 10^{-17} \text{ N}$
(b) $(6.4\hat{i} + 1.6\hat{j}) \times 10^{-17} \text{ N}$
(c) $4.0 \times 10^{10} \text{ m/s}^2$
(d) $7.3 \times 10^{13} \text{ m/s}^2$